

A Piezoelectric Implant to Promote Cardiac Healing through Motion-Driven Electrical Stimulation

Capstone Project #11 Report

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Executive Summary

Cardiovascular diseases, particularly those caused by myocardial infarction, remain a leading cause of death globally. Infarction-induced damage results in scar tissue formation, compromising heart function and reducing patients' quality of life. While current regenerative therapies, such as stem cell treatments, show promise, they face limitations including low cell engraftment, immune rejection, and poor tissue integration. There is a clear need for innovative, effective approaches to cardiac repair.

This project aimed to validate a novel polyvinylidene difluoride (PVDF) implant designed to support cardiac tissue regeneration through strain-induced electrical stimulation. To achieve this, we developed a custom-built, automated stretcher system that mimics the cyclic mechanical environment of the beating heart, enabling *in vitro* testing of the implant's ability to generate voltage and promote cardiomyocyte proliferation. The platform was engineered to meet specific functional requirements: use of biocompatible and stable materials, cyclic mechanical stimulation, compatibility with incubator conditions, support for parallel testing, and generation of physiologically relevant mechanical stress. Together, these features ensure reproducibility, scalability, and relevance for future translational applications.

Our solution represents a promising step forward in cardiac regenerative medicine—providing a minimally invasive, scalable, and energy-autonomous strategy to repair damaged heart tissue and improve long-term patient outcomes

Key words: Piezoelectric Implant, Cardiac Regeneration, Electrical Stimulation, Myocardial Infarction

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Introduction

1.1. Heart Disease: A Global Health Crisis

Cardiovascular diseases (CVDs) are the leading cause of death worldwide, responsible for approximately 17.9 million deaths each year, accounting for nearly a third of all global deaths (WHO, 2024). Among these, ischemic heart disease and stroke are the most prevalent, affecting millions of people across all demographics (Chong et al., 2024). Despite significant advancements in medical treatments, CVDs continue to challenge healthcare systems worldwide, not only due to their long-term consequences on heart function. Among the most severe outcomes of heart disease is myocardial infarction, also commonly known as heart attack, which leads to irreversible damage to the heart muscle (WHO, 2024).

1.1.1 Biological Mechanisms of Heart Damage

CVDs are primarily caused by the blockage of coronary arteries, which supply oxygen-rich blood to the heart muscle (John Hopkins Medicine, 2019). This blockage typically results from the rupture of plaques, mainly cholesterol, that form deposits in the arteries over time. The rupture leads to the formation of a clot that obstructs blood flow, causing ischemia—lack of oxygen—within the affected region of the heart. Without proper oxygenation, the heart muscle cells, or cardiomyocytes, begin to die, and the heart's ability to pump blood is compromised, counting up to 37% function reduction [Figure 1.1] (Mayo Clinic, 2023) (Mount Elizabeth Hospitals, 2021)(Chiong et al., 2011).

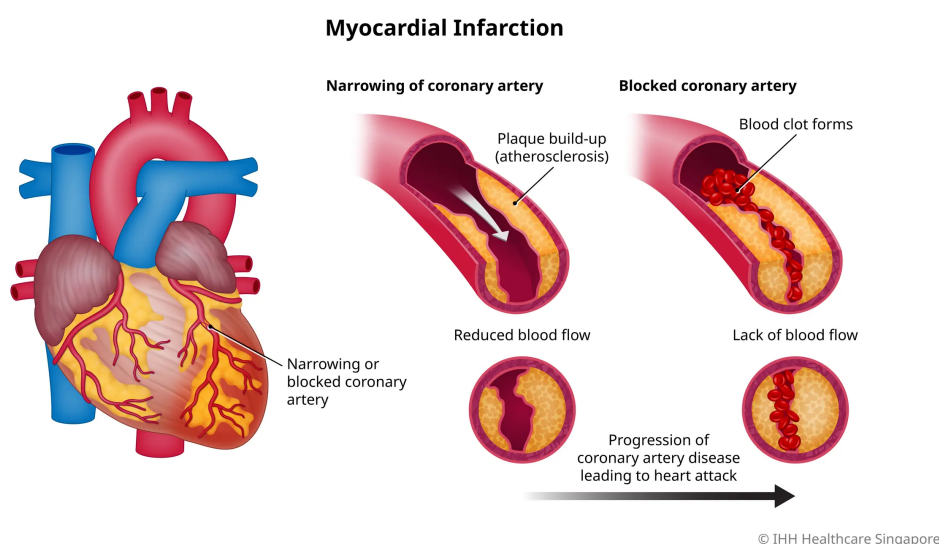


Figure 1.1: Diagram representing coronary artery blockage during myocardial infarction (Mount Elizabeth Hospitals, 2021)

This ischemic event triggers a cascade of cellular damage, ultimately leading to the death of cardiomyocytes in the affected area. Unlike other tissues, the heart has a limited regenerative capacity, meaning that the lost muscle cells are not replaced (Richardson et al., 2015). Instead, the body responds by

forming fibrotic scar tissue in the infarct region. While this scarring prevents structural collapse, it lacks the contractile properties of healthy myocardium, weakening the heart's ability to pump blood efficiently (Lodrin and Goumans, 2021). Over time, the presence of non-functional scar tissue disrupts the electrical and mechanical integrity of the heart, increasing the risk of arrhythmia, heart failure, and further deterioration of cardiac function.

1.1.2 Current Post-Heart Attack Treatments are Limiting

Existing therapies for myocardial infarction primarily focus on restoring blood flow and managing symptoms rather than reversing tissue damage. Standard treatments include pharmacological interventions such as beta-blockers, ACE inhibitors, and anticoagulants to reduce strain on the heart and prevent further clot formation. In severe cases, surgical procedures like coronary artery bypass grafting (CABG) or percutaneous coronary interventions (PCI) are used to restore circulation (Sweis, R., and Jivan, A., 2018). However, while these approaches improve survival rates (35.6% improvement) (Weintraub et al., 2003), they do not address the underlying issue of infarct-induced cardiomyocyte loss.

To promote tissue regeneration, researchers have explored stem cell-based therapies as a potential solution. Mesenchymal stem cells (MSCs), induced pluripotent stem cells (iPSCs), and cardiac progenitor cells have been investigated for their ability to differentiate into cardiomyocytes and integrate into damaged myocardium (Chu, Jiang, and Xiao, 2024)(Zhang et al., 2019). However, clinical outcomes have been inconsistent due to low engraftment rates, immune rejection, and limited functional integration of transplanted cells into the native cardiac tissue. Additionally, while biomaterial scaffolds and engineered heart tissue offers structural support, they struggle to fully replicate the native myocardial microenvironment and electrical conductivity required for synchronized contractions.

These limitations highlight the urgent need for alternative regenerative strategies that can directly stimulate endogenous cardiac repair mechanisms.

1.2. Heart Regeneration

In the search for innovative approaches to cardiac repair, we turned to nature for inspiration. Certain vertebrates, such as Teleost fish, possess a remarkable ability to regenerate heart tissue after injury. These species can effectively replace lost cardiomyocytes and restore full cardiac function, preventing long-term scarring. Unfortunately, this regenerative capacity is largely absent in adult mammals, including humans, but what if we could externally stimulate proliferation?

1.2.1 Electrically Induced Regeneration

Electrical stimulation (ES) has emerged as a promising strategy for promoting cardiomyocyte proliferation, differentiation, and functional maturation (Dai, Mu, and Zhou, 2021). The heart is an electromechanical organ, where bioelectrical signals coordinate contraction and maintain tissue homeostasis. By mimicking these endogenous electrical cues, ES has the potential to activate key cellular pathways that support myocardial repair and regeneration [Figure 1.2].

Electrical stimulation (ES) is emerging as a frontier in cardiac tissue regeneration due to its ability to directly influence cardiomyocyte behavior and promote functional tissue repair. At the cellular level, ES activates voltage-gated ion channels—particularly calcium (Ca^{2+}) and potassium (K^+) channels—triggering intracellular signaling cascades that drive cardiomyocyte proliferation, structural maturation, and contractility (Bers, D. M., 2002). Calcium influx, initiated by membrane depolarization, activates key effectors like CaMKII and RyR2, enhancing gene expression, sarcomere organization, and synchronized contractions (Tandon et al., 2011).

Beyond ionic effects, ES upregulates critical cardiac transcription factors such as GATA4, NKX2.5, and MEF2C, and activates pro-survival pathways like ERK1/2 and Akt (Bruneau, B. G., 2013). It also enhances intercellular connectivity by increasing connexin-43 (Cx43) expression, crucial for maintaining coordinated beating. Moreover, ES influences extracellular matrix remodeling by promoting deposition of fibronectin and collagen IV, providing structural integrity for new tissue. This electro-mechanical synergy closely mimics the native myocardial environment, making ES a potent stimulus for both cellular and tissue-level regeneration (Riehl et al., 2012).

The benefits of external electrical stimulation

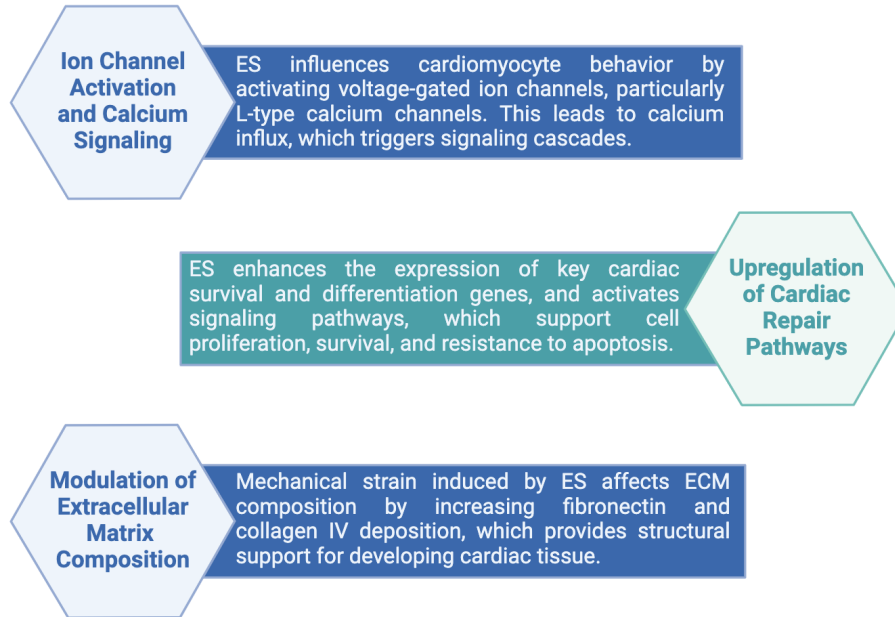


Figure 1.2: Diagram representing the benefit of electrical stimulation on the cardiac tissue and extracellular matrix (Bers, D. M., 2002)(Bruneau, B. G., 2013)(Riehl et al., 2012)

By harnessing these mechanisms through a self-powered piezoelectric implant, we aim to translate the regenerative potential of ES into a clinically viable therapy for myocardial infarction.

1.3. A Piezoelectric Implant as a Novel Therapeutics Device

To address the limitations of current cardiac regenerative therapies, we propose a novel approach: a piezoelectric implant that leverages the heart's mechanical motion to generate bioelectrical stimulation. Piezoelectric materials are unique as they convert mechanical stress into electrical energy, a property that has gained increasing attention for biomedical applications (Wu et al., 2024). Among these materials, polyvinylidene difluoride (PVDF) stands out as a biocompatible, flexible, and stable piezoelectric polymer—qualities that make it particularly well-suited for cardiac implantation (ForeTec Digital, 2024). As a semi-crystalline polymer, PVDF can form multiple polymorphic phases (α , β , and γ), with the β -phase being the most desirable due to its high dipole moment and superior piezoelectric properties.

1.3.1 Hypothesis

The implant design consists of a thin PVDF film applied on infarcted cardiac tissue. As the heart contracts and relaxes, the film undergoes periodic deformation, generating a localized electrical potential through its piezoelectric properties. This strain-induced voltage propagates through the myocardial extracellular matrix, creating an electrically active microenvironment that mimics the physiological bioelectrical signals essential for cardiac repair. By providing continuous, self-sustained electrical stimulation, the PVDF-based implant is expected to promote cardiomyocyte proliferation, enhance ECM remodeling, and encourage functional tissue regeneration in infarcted regions [Figure 1.3].

Unlike external electrical stimulation devices, which require invasive implantation and external power sources, our piezoelectric implant operates autonomously, eliminating the need for batteries or wired connections. This not only reduces the risk of complications associated with foreign implants but also ensures a seamless integration with native cardiac tissue. If successful, this approach could provide a groundbreaking therapeutic strategy for patients suffering from myocardial infarction, offering a self-powered, bioelectrically active scaffold to reverse the detrimental effects of cardiac tissue scarring.

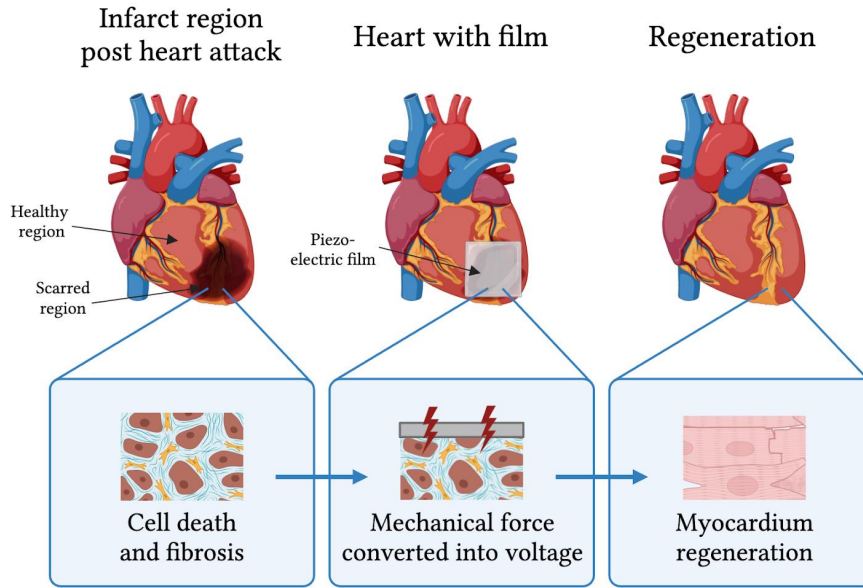


Figure 1.3: Diagram illustrating our hypothesis in three stages: the initial state before applying the piezoelectric implant, the implantation of the PVDF film, and the final stage with restored cardiac tissue.

1.3.2 Current Approaches

In previous studies, researchers manually applied mechanical stress and strain to piezoelectric films and observed their effects on cardiomyocyte behavior. These experiments revealed a trend of increased beating frequency in response to the electrical stimulation induced by the film. The promising outcomes from these preliminary investigations laid the foundation for our capstone project.

However, manual application of strain is limited by low reproducibility, lack of precise control, and an inability to sustain physiologically relevant, continuous stretching. To address these limitations and better simulate the mechanical environment of the heart, an automated stretching system needs to be used. This system would enable consistent, cyclic mechanical stimulation that mimics the natural motion of the myocardium, providing a more physiologically accurate model for *in vitro* testing.

Furthermore, the automated stretcher would allow for fine control over strain parameters such as frequency, amplitude, and duration, ensuring that the mechanical input is both repeatable and tunable. This not only improves experimental reproducibility but also enables systematic testing of the implant under various conditions.

Existing Device Limitations

Commercially available stretch systems, such as the FLEXCELL Tension System, are designed primarily for stretching cell monolayers cultured on flexible membranes, not for mechanically deforming thin piezoelectric films like PVDF. These systems are often cost-prohibitive, support only a single sample per run, and may lack the precision needed to deliver cyclic strain at physiologically relevant frequencies such as 1 Hz. Moreover, most commercial systems are bulky and not compatible with use inside standard incubators, limiting their ability to provide continuous mechanical stimulation during cell culture.

1.3.3 Goals of Our Project

The goal of this project was to validate our piezoelectric implant using a custom-built automated stretcher system that mimics the heartbeat, enabling *in vitro* testing of its ability to generate voltage and promote cardiomyocyte proliferation. Using this system, which replicates the heart's mechanical environment, we aim to quantify the strain-induced voltage output and establish the optimal electrical conditions needed to stimulate cardiomyocyte proliferation. Additionally, we performed *in vitro* experiments with cell cultures to evaluate the biological effects of the generated electrical stimulation

on cardiac tissue regeneration. Finally, we conducted a thermal and mechanical characterization of the PVDF film to determine how these properties influence its performance as a cardiac regenerative therapy.

To improve experimental reproducibility through automation and standardization, our device was designed with specific functional criteria in mind. It must:

1. Be composed of biocompatible materials.
2. Utilize a stable material.
3. Apply continuous, cyclic stretching to mimic natural heart motion.
4. Be suitable for *in vitro* testing and compact enough to fit within a standard incubator.
5. Enable testing of multiple experimental conditions in parallel.
6. Generate physiologically relevant stress and strain levels.

Together, these design goals ensure our platform not only supports reliable validation of the piezoelectric implant but also lays the groundwork for future translational studies, ultimately advancing the feasibility of piezoelectric materials for myocardial repair.

Methods and Approach

2.1. Characterization

Our project focuses on a piezoelectric film, making its characterization critical to understand key properties. The piezoelectric material utilized in this project was acquired from Polyk Technologies (SKU: PVDF-Al-P0007).

2.1.1 Thermal Characterization Using Differential Scanning Calorimetry

To ensure reliable and physiologically relevant operation of the device, it is essential to understand the thermal properties of its core functional materials—particularly the PVDF film used for electromechanical transduction. PVDF's piezoelectric properties are inherently sensitive to temperature, and its performance can degrade or change significantly near critical thermal transition points. Thermal analysis serves as a foundational step in validating the reliability and biocompatibility of the PVDF film within a controlled *in vitro* experiment. Accurate thermal profiling also ensures that stress and strain outputs remain consistent over time, supporting experimental reproducibility and enabling automation. Ultimately, it mitigates the risk of material fatigue or depolarization that could compromise cyclic stretching fidelity, especially during long-term studies.



Figure 2.1: Differential scanning calorimetry machine with samples.

To assess the thermal properties of the PVDF film, Differential Scanning Calorimetry (DSC) was used [Figure 2.1]. DSC is a technique that measures the heat flow associated with the thermal transitions of a material as it is heated or cooled, providing valuable information about phase changes, crystallinity, and the thermal stability of the polymer. This technique is particularly important for evaluating PVDF, as its piezoelectric properties are highly dependent on the crystalline phases present in the material.

Approximately 3 mg of the PVDF film was cut into small pieces and placed in an aluminum crucible. The samples were heated from 25°C to 500°C at a rate of 10°C per minute under a nitrogen atmosphere to minimize oxidation. The heat flow was recorded, and the resulting thermographs were analyzed to identify key thermal transitions, including the glass transition temperature (T_g), the melting temperature (T_m), and the degradation temperature.

This data is essential for ensuring the film's performance in the piezoelectric cardiac stimulation system, as it confirms that the material exhibits the appropriate thermal stability and crystallinity for optimal piezoelectric behavior.

2.2. Stretch System

In order to replicate the mechanical stimulus experienced by cardiac tissue *in vivo*, a stretch system was developed to cyclically deform the piezoelectric PVDF film at a frequency of 1 Hz. The system is composed of several integrated components: a 6-well plate, a PDMS film, a sealed chamber, and an external pump connected to a microcontroller-controlled circuit that cyclically inflates the membrane. Together, these elements ensure that the film undergoes mechanical stress closely resembling the natural beating frequency of the human heart, providing a physiologically relevant mechanical environment (5-15% strain). This modular setup allows for continuous stretching, fits within standard incubators for *in vitro* testing, enables testing under multiple conditions, and improves reproducibility—ensuring that all functional goals for the device are met.

2.2.1 Design of the Stretch System

The stretch system consists of a sealed plastic vacuum chamber with an inlet for air flow control. The chamber was designed using SolidWorks to ensure precise dimensions to fit a multi-well cell culture plate. The inlet of the chamber allows for controlled inflation of a poly(dimethylsiloxane) (PDMS) membrane that would sit on top and comes in contact with the PVDF film.

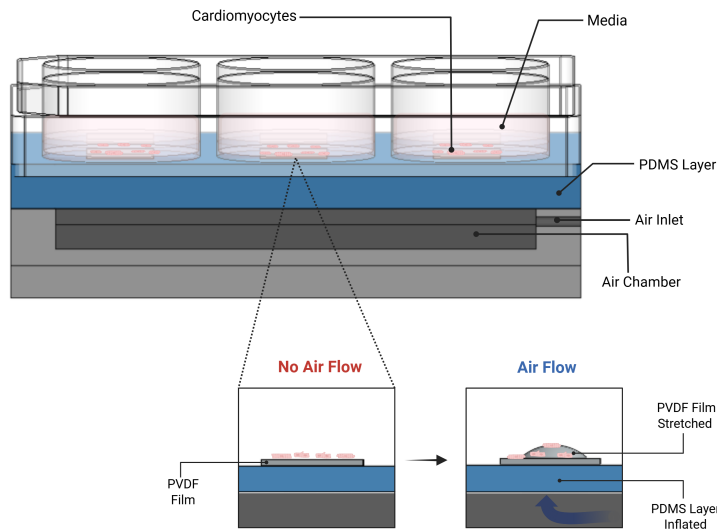


Figure 2.2: Diagram of a cyclic stretch system for cardiomyocyte culture, utilizing a PVDF film secured to a modified 6-well plate. The system operates with a plastic chamber, where controlled inflation of the PDMS layer induces mechanical stretching of the PVDF film. This mechanical force is applied to cardiomyocytes cultured on the film, generating electrical signals through the piezoelectric effect.

A mold for casting the PDMS layer was designed in Solidworks and 3D printed using polylactide (PLA) material. The PDMS Sylgard 184 was prepared by mixing in a 10:1 ratio of base to crosslinker and poured into the printed mold. To facilitate easy removal of the membrane, the mold was covered with saran wrap before pouring the PDMS mixture. After pouring, the PDMS mixture was degassed using a desiccator for one hour to remove air bubbles. The mold was then placed in an oven at 40°C overnight for complete curing. Once solidified [Supplementary Figure A.1], the PDMS layer was extracted from the mold and glued to the plastic vacuum chamber using PDMS [Supplementary Figure A.2]. The entire part was placed in the oven again at 40°C for proper adhesion. The tubing for air flow was attached by gluing it into the inlet using PDMS, ensuring a sealed connection. This setup allows for cyclic inflation in a semi-spherical manner, applying controlled mechanical deformation to a PVDF film that is fixed to the bottom of a milled cell culture well plate. The device is designed to exert stress on the piezoelectric film, straining the film in a way that enables it to emit an electric charge into a cardiomyocyte cell culture grown on top of the PVDF film [Figure 2.2].

2.2.2 Integration with Cell Culture Well Plate

A standard 6-well plate was modified to facilitate direct mechanical interaction with the PVDF film. A 10 mm hole was drilled into the center of each well using a mill machine (G0935 1-1/2 HP Mill/Drill) [Supplementary Figure A.3]. The holes were sanded to prevent rough edges that could affect adhesion. The PVDF films were cut into 15 mm squares to ensure proper coverage over the drilled holes. To examine the influence of piezoelectric properties on cellular response, both poled and non-poled PVDF film were used. Each respective film was glued to the bottom of the plate using PDMS Sylgard 184. To ensure a secure bond, the modified well plates were placed in an oven at 40°C overnight for curing. After curing, the plates were filled with deionized water to confirm that they were properly sealed and there was no leakage. Following the sealing test, the plates were sterilized to prepare them for cell culture experiments.

2.2.3 Cyclic Pump System

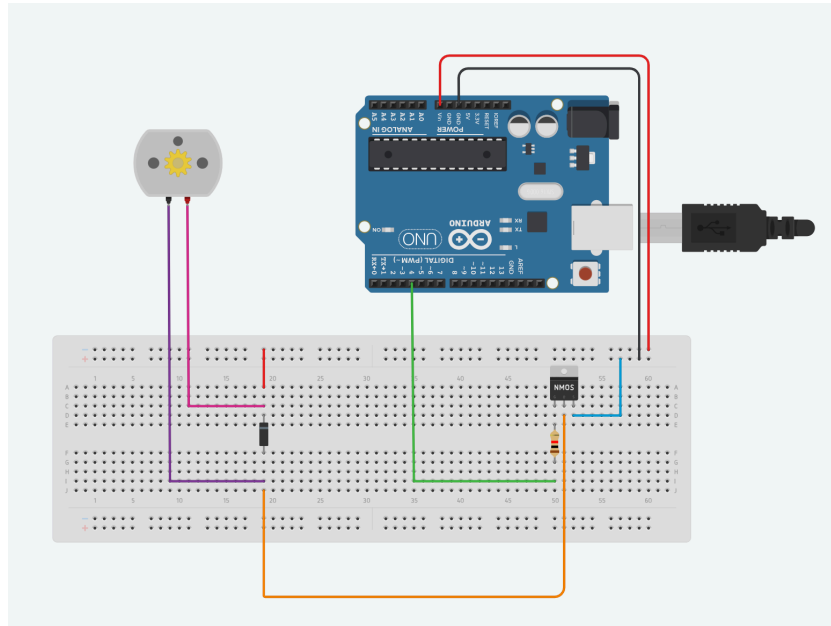


Figure 2.3: Breadboard circuit layout for automated solenoid valve control using an Arduino Uno R4 and MOSFET. This hardware configuration enables digital control of pneumatic actuation at 1 Hz, facilitating cyclic mechanical stimulation in the experimental setup (Chris @ BCR, 2015).

To replicate the mechanical environment of the beating heart, we developed an automated cyclic pump system designed to stretch the PVDF film at a physiologically relevant frequency of 1 Hz (60 beats per minute). This system utilizes a 12V pneumatic solenoid valve to regulate the flow of pressurized air into a 3D printed air chamber, thereby inflating and deflating a PDMS membrane. As the PDMS layer

expands, it transmits mechanical strain to the PVDF film secured to the modified 6-well plate. This inflation and deflation cycle reproduces the periodic stretching that cardiac tissues naturally undergo, enabling *in vitro* studies of piezoelectric film behavior under physiologically relevant mechanical stimuli.

An Arduino Uno R4 microcontroller was programmed to generate a digital pulse signal at 1 Hz, corresponding to the desired stretch frequency. This signal is sent to a metal-oxide-semiconductor field-effect transistor (MOSFET), which acts as an electronic switch to control the solenoid valve. The MOSFET allows the high current required by the solenoid to be controlled safely by the low-power Arduino output. The system circuit, as illustrated in Figure 2.3 and Supplementary Figure A.4, consist of several key components. A N-channel MOSFET is used to switch the solenoid valve on and off in response to the Arduino signal. A 1 k Ω resistor is placed between the Arduino's digital output pin and the MOSFET gate to limit the gate current and protect the microcontroller from potential damage. A flyback diode is connected across the solenoid terminals to suppress voltage spikes generated by the inductive load during switching events. The solenoid is powered by an external 12V power supply, as the Arduino cannot provide sufficient current. The integration of microcontroller-based timing with pneumatic action enables repeatable and continuous stretching cycles [Figure 2.4]. This setup allows for cell culture in parallel with PVDF films under dynamic strain conditions.

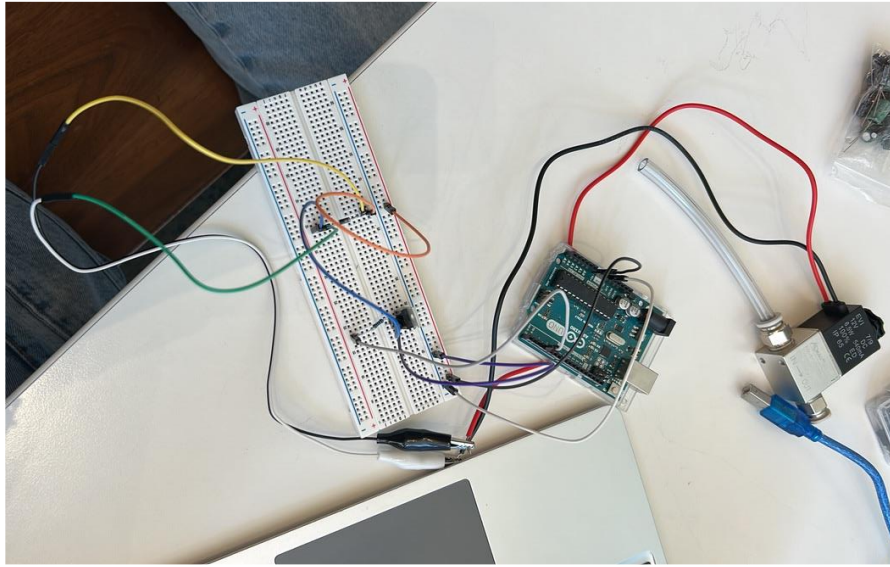


Figure 2.4: Breadboard circuit assembled for automated solenoid valve control using an Arduino Uno R4 and MOSFET (Chris @ BCR, 2015).

2.2.4 Overall System Configuration

The stretch system integrates a piezoelectric PVDF film with a custom-built pneumatic stretch platform to simulate cardiac-like mechanical stimulation. The PVDF film is mounted on a modified 6-well plate and cyclically deformed by a PDMS membrane, which inflates in response to pressurized air controlled by a solenoid valve and Arduino circuit operating at 1 Hz. This setup mimics the mechanical environment of the heart and is compact enough to fit inside a standard incubator for continuous cell culture testing. The modular design enables parallel experiments and supports evaluation of biocompatibility and material behavior under dynamic strain conditions.

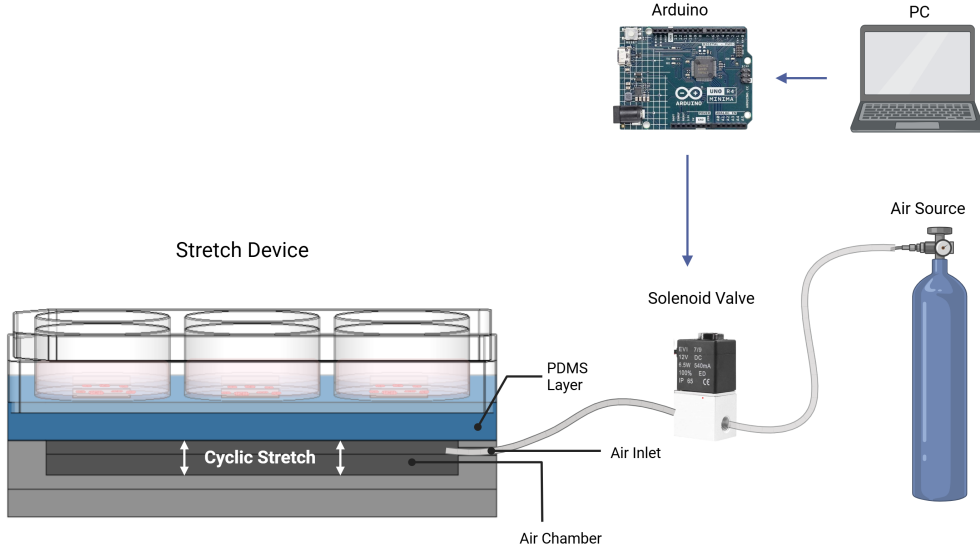


Figure 2.5: Schematic of a pneumatic stretch device used to apply cyclic strain to cardiomyocytes cultured on a flexible PDMS membrane. Compressed air, controlled by a solenoid valve, inflates an air chamber beneath the membrane to simulate mechanical stimulation similar to that of a beating heart.

2.2.5 Testing the Inflation-Deflation Response

PDMS testing

Once the overall system was built, we had to evaluate the inflation-deflation response and deformation behavior of the PDMS membrane. A 1 mm diameter tube was connected to the bottom chamber, while a small incision in the chamber seal acted as a passive outlet to facilitate deflation, enabling cyclic actuation. The inflation of the membrane was driven by regulated airflow, and the minimum volumetric flow rate required for observable inflation was determined.

To estimate the dynamic pressure required to initiate membrane deformation, Bernoulli's equation for incompressible flow was used:

$$P = \frac{1}{2} \rho v^2 \quad (2.1)$$

where P is the pressure in pascals (Pa), ρ is the air density (taken as 1.225 kg/m^3), and v is the velocity of air in m/s, which can be calculated from the flow rate Q and cross-sectional area A of the tube using:

$$A = \pi r^2 \quad (2.2)$$

$$v = \frac{Q}{A} \quad (2.3)$$

For a tube with radius $r = 0.5 \text{ mm}$ and a flow rate of $Q = 20 \times 10^{-6} \text{ m}^3/\text{s}$, the velocity and resulting dynamic pressure were calculated.

To quantify the displacement of the membrane, the inflation was modeled as a spherical cap. The vertical amplitude h of the dome was measured from the undeformed membrane surface to the peak of the inflated membrane. Given the known membrane radius r , the displaced volume V was calculated using:

$$V = \frac{1}{6} \pi h (3r^2 + h^2) \quad (2.4)$$

This characterization method established a baseline for assessing the mechanical behavior of the stretch system and provided the foundation for later electromechanical coupling experiments.

PVDF Strain Testing

To evaluate the mechanical strain induced to the PVDF film by our cyclic stretching system, we employed a visual tracking method using markers and image analysis. Two small ink dots were placed on the surface of the PVDF film approximately 0.11 cm apart in the unstretched state. The stretch system was then operated, and high-resolution images were captured at both the relaxed and maximum inflation positions of the PDMS membrane.

The change in distance between the markers was measured using ImageJ, an open-source image analysis software. This enabled calculation of engineering strain using the standard formula:

$$\text{Strain} = \frac{\Delta L}{L_0} = \frac{L - L_0}{L_0} \quad (2.5)$$

where L_0 is the initial distance between markers, and L is the distance when the film is fully stretched. This method provided a non-invasive and accessible means of quantifying strain to ensure that the applied mechanical stimulus was physiologically relevant.

2.3. Cell Culture

To evaluate the biocompatibility of PVDF substrates and their suitability for electro-mechanical stimulation, cardiomyocytes were cultured under two experimental conditions, a cytotoxicity assay to assess viability across PVDF substrate types, and a stretching experiment to investigate the cell response under mechanical strain. All cells were supplied in suspension and seeded directly. Culture protocols were followed based on guidelines provided by the cell supplier.

2.3.1 Plate Preparation for Cell Culture

To prepare the wells for cell seeding, all materials were first sterilized. Well plates with attached films were sterilized using a Stratalinker UV crosslinker on the "crosslink" setting to eliminate biological contaminants. Well plates were then wiped with 70% ethanol and rinsed with sterile deionized water. For experiments involving cell adhesion, extracellular matrix (ECM) coating was applied to enhance attachment. A solution of Matrigel was prepared by diluting a 10 mg/mL Matrigel stock at a 1:5 ratio in cold PBS. This was further diluted to a final concentration of 1:100 in PBS, and 3 mL of ECM solution was added to each well. Plates were left on a room temperature shaker for 1 hour to ensure uniform coating.

2.3.2 Cell Types and Media

Two cell types were used for experiments: primary mouse myoblasts and human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs). Myoblasts were cultured in F-10 Nutrient Mixture supplemented with 20% Hyclone bovine growth serum, 1% penicillin-streptomycin, and 10 ng/mL basic Fibroblast Growth Factor (bFGF). hiPSC-CMs were initially seeded in EB20 medium supplemented with ROCK inhibitor (RI) to support attachment and survival. After 24 hours, the medium was replaced with RPMI-1640 + B27 supplement, which supports long-term culture and cardiomyocyte maturation. All cultures were maintained at 37°C in a humidified incubator with 5% CO₂, and media was changed every other day.

2.3.3 Myoblast Cytotoxicity Assay

Initial biocompatibility testing of PVDF substrates and adhesives was performed using myoblasts. Four groups were tested: poled PVDF, non-poled PVDF, aluminum-coated PVDF, and a control group (no film). Cells were seeded at 100,000 cells per well in ECM-coated 6-well plates. After seeding, morphology and attachment were monitored using bright-field microscopy to identify any cytotoxic effects of the film materials or adhesives.

2.3.4 HiPSC-CM Cytocompatibility and Stretching Experiments

For the first cardiomyocytes experiment, 1.5 million cells were divided among three groups: (1) poled PVDF film, (2) non-poled PVDF film, and (3) an unmodified with no film control well. All wells were coated with ECM, and after 24 hours in EB20 + RI, media was switched to RPMI + B27.

In a subsequent experiment, the full 1.5 million cells were seeded into a single well containing poled PVDF film. This setup aimed to increase local cell density and improve intercellular connectivity. The well was incorporated into a custom cyclic stretching device, described in Section 2.2.3. Stretching was initiated after the media switch and continued for 24 hours using a ramp-up protocol to minimize detachment. Imaging was performed before stretching and once daily for three days to observe morphological changes [Figure 2.6].

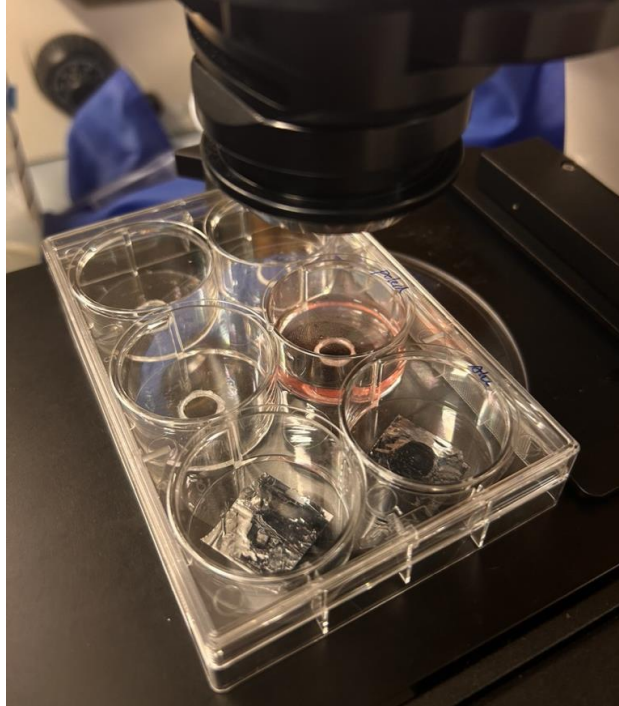


Figure 2.6: Well plate analyzed under microscope.

3

Results

As previously outlined, the device was engineered to meet a defined set of functional requirements critical for its intended application in cardiac tissue modeling. It was constructed exclusively from biocompatible materials to ensure cellular compatibility and prevent cytotoxic effects during extended *in vitro* experimentation. The system was designed to deliver continuous, cyclic mechanical deformation to replicate the dynamic strain experienced by myocardial tissue. Dimensional constraints were considered to ensure compatibility with standard incubators, allowing integration into routine cell culture workflows. To facilitate high-throughput analysis and experimental efficiency, the platform supports parallel testing of multiple conditions. Moreover, the mechanical output of the device was calibrated to produce physiologically relevant stress and strain amplitudes reflective of native cardiac biomechanics. Finally, the system architecture prioritized automation and operational standardization to enhance reproducibility and minimize user-dependent variability.

3.1. Composed of Biocompatible Materials

The performance of the device was assessed through a series of *in vitro* experiments to evaluate material biocompatibility, cell viability, and electromechanical stimulation outcomes. Initial cytotoxicity testing was conducted using primary mouse myoblasts seeded onto four surface conditions: poled PVDF, non-poled PVDF, aluminum-coated PVDF, and control wells. All surfaces were coated with ECM to promote adhesion. Myoblasts seeded at a density of 100,000 cells per well exhibited typical spindle-shaped morphology with no signs of cytotoxicity, including cell rounding or detachment, across all conditions, indicating that the PVDF substrates and adhesives used to secure the films were biocompatible. Following this, human iPSC-derived cardiomyocytes were cultured on three conditions—poled PVDF, non-poled PVDF, and control—by dividing a total of 1.5 million cells among the three wells. While cells displayed generally healthy morphology in the poled and control groups, the low seeding density led to poor confluency, likely impairing the formation of intercellular connections required for synchronized beating.

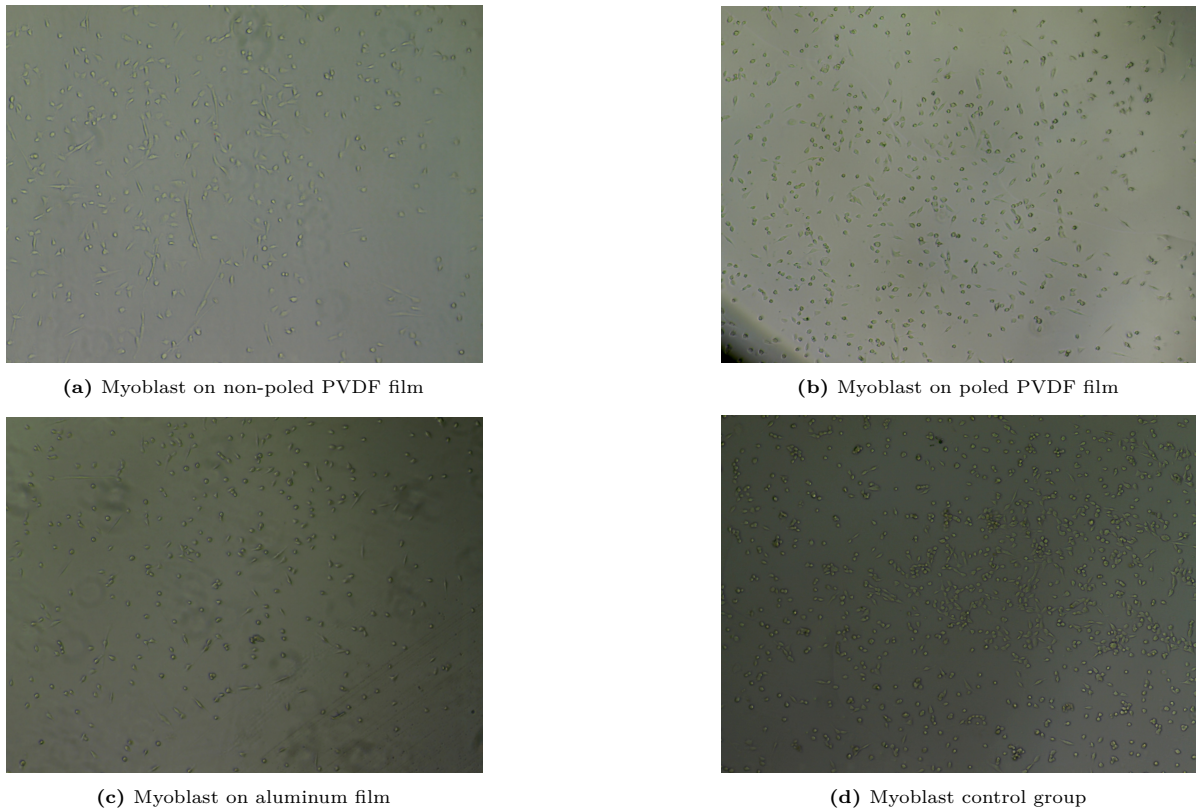


Figure 3.1: Bright-field microscopy images of myoblasts on poled, non-poled PVDF film, aluminum film and control

To address the confluency issue, a subsequent experiment seeded all 1.5 million cardiomyocytes into a single well with poled PVDF film to increase cell density. Although improved attachment was observed, the large surface area of the 6-well plate still prevented the formation of a complete syncytium, and no spontaneous contractions were detected. Cyclic mechanical stretching was initiated after 5 days in culture using a ramp-up protocol, and the device successfully operated within an incubator environment for 24 hours over the course of 3 days. Microscopy at 24, 48, and 72 hours post-stretch confirmed that cells remained adherent with stable morphology throughout the stimulation period [Figure 3.2]. These findings demonstrate that the PVDF materials and adhesive system are compatible with both myoblasts and cardiomyocytes under static and mechanically stimulated conditions. However, the limited availability of cardiomyocytes prevented functional maturation, underscoring the need for future experiments with higher seeding densities or smaller culture areas. Additionally, delaying mechanical stimulation until spontaneous beating is observed could allow for more effective evaluation of contractile changes and cardiomyocyte maturation. Overall, the system shows strong promise for application in long-term electromechanical studies that are adaptable for cardiac and muscle regeneration research.

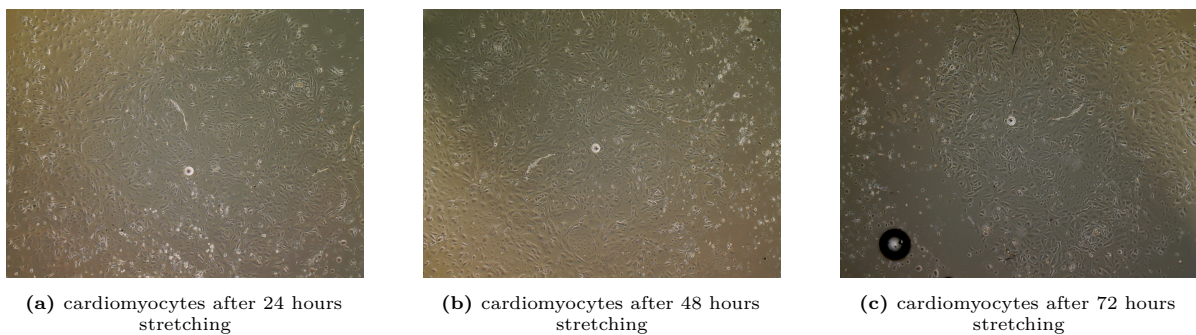


Figure 3.2: Bright-field microscopy images of cardiomyocytes on poled PVDF film at different time points. Cells show stable attachment and morphology over time.

3.2. Utilize a Stable Material

Differential Scanning Calorimetry (DSC) was employed to analyze the thermal properties of three different PVDF samples: non-poled and non-coated PVDF, poled and non-coated PVDF, and poled and aluminum-coated PVDF. All samples were analyzed under first heat. The purpose of this analysis was to assess how poling and aluminum coating influence the thermal transitions of PVDF, including its melting behavior, crystallization, and degradation. Understanding these thermal properties is critical for evaluating the material's stability, phase composition, and suitability for applications in piezoelectric devices.

The thermal transitions observed in DSC provide insights into the phase composition and structural stability of the material under different conditions. Poling, which involves applying an electric field to align dipoles within the polymer, is known to enhance the formation of the β -phase, while aluminum coating may affect thermal conductivity and interaction with the polymer matrix. By analyzing all three samples, we establish a baseline for the untreated PVDF and observe modifications introduced by poling and coating. This information is crucial for optimizing the processing conditions of PVDF-based piezoelectric materials and ensuring their stability in practical applications.

The DSC was ran from 25°C to around 500°C. The Glass Transition Temperature (T_g) of PVDF is typically observed around -35°C to -40°C (Dallaev et al., 2022), indicating that the transition was outside the scanned temperature range.

3.2.1 Control: Non-coated and non-poled PVDF Piezoelectric Film

The DSC analysis of the non-poled, non-coated PVDF sample reveals distinct thermal transitions characteristic of the polymer [Figure 3.3].

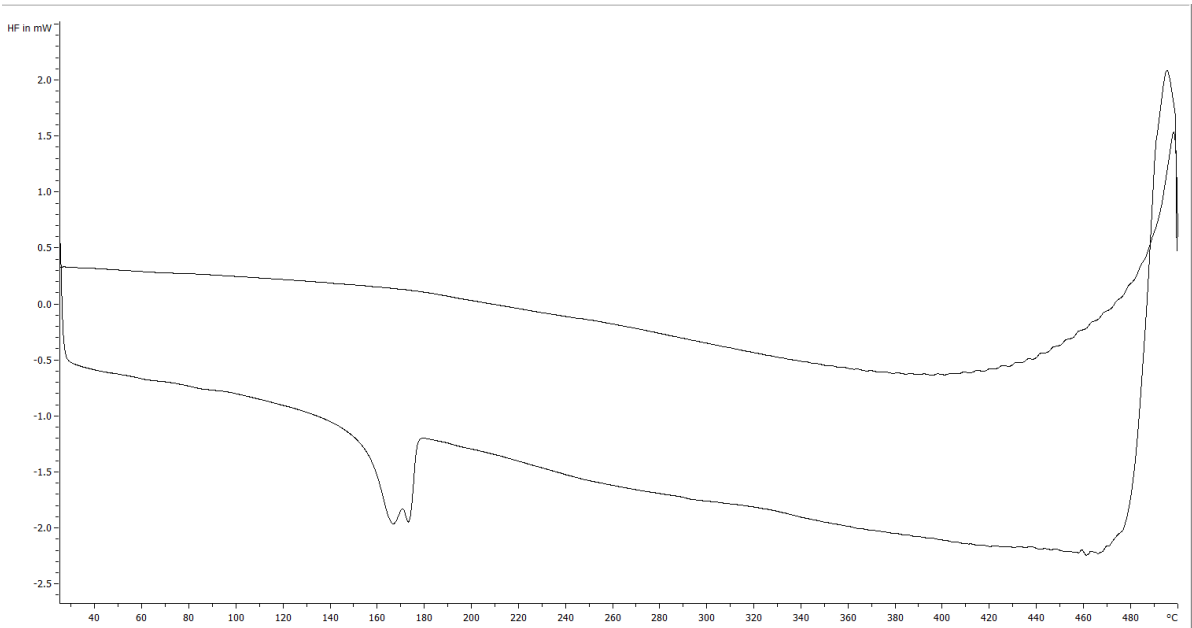


Figure 3.3: DSC thermograph of the non-coated non-poled PVDF piezoelectric film.

Melting Behavior and Crystallinity: The melting peak appears as a bimodal feature between approximately 160°C and 180°C, which is consistent with the crystalline phase transitions of PVDF. The first peak around 160°C may correspond to the melting of the β -phase, while the second, more prominent peak near 175–180°C is likely associated with the melting of the α -phase. This suggests the presence of a mixed crystalline structure, even though the sample has not undergone electrical poling to intentionally induce β -phase formation. The β -phase may have formed in small quantities due to processing conditions.

Thermal Degradation: At higher temperatures, thermal degradation begins at approximately 420°C, indicated by a sharp exothermic drop in the heat flow curve. This degradation temperature suggests that the polymer maintains its thermal stability up to this point, which aligns with reported degradation profiles for PVDF.

3.2.2 Non-coated Poled PVDF Piezoelectric Film

The DSC thermograph [Figure 3.4] of the non-coated PVDF piezoelectric film exhibits key thermal transitions characteristic of PVDF, including its melting temperature (T_m) and degradation onset.

Melting Behavior and Crystallinity: A sharp endothermic peak was observed at approximately 175°C, corresponding to the T_m of the crystalline phase of PVDF. In this case, the peak is not bimodal, indicating the presence of one unique phase: β -phase. This value aligns with previously reported melting temperatures of PVDF, which range between 170°C and 180°C, depending on the polymer's crystallinity and processing conditions (Alfa Chemistry, 2024). The presence of a well-defined melting peak suggests a relatively high degree of crystallinity in the non-coated but poled PVDF sample.

Thermal Degradation: A broad exothermic transition occurs between 420°C and 480°C, indicating the onset of thermal degradation. PVDF is known to undergo thermal decomposition at temperatures exceeding 400°C, primarily due to the breakdown of its polymer backbone and the release of HF (hydrofluoric acid) (Saxena and Shukla, 2021). The presence of a distinct degradation peak confirms the polymer's thermal stability up to approximately 400°C.

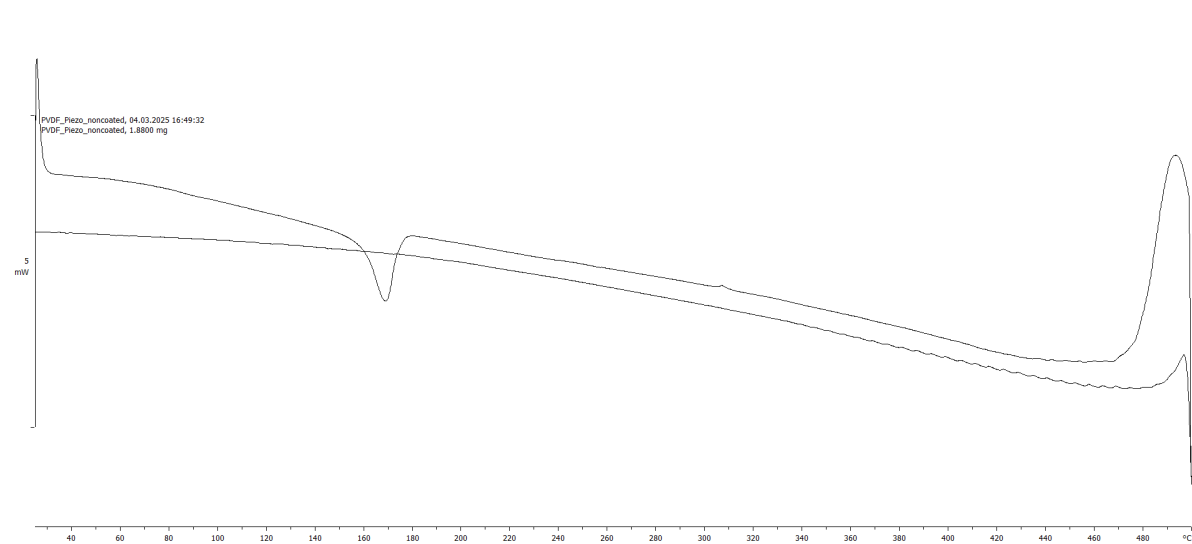


Figure 3.4: DSC thermograph of the non-coated poled PVDF piezoelectric film.

The DSC analysis confirms the expected thermal transitions of PVDF, demonstrating a sharp melting peak at 175°C, and thermal stability up to 400°C before degradation begins. The results indicate that the non-coated poled PVDF piezoelectric film possesses a high degree of crystallinity, which is essential for its piezoelectric performance.

3.2.3 Aluminum-Coated Poled PVDF Piezoelectric Film

The DSC thermograph of the aluminum-coated poled PVDF piezoelectric film [Figure 3.5] reveals the material's thermal transitions, including melting behavior, crystallinity, and thermal degradation onset. Comparing this sample with the non-coated poled PVDF film, the presence of the aluminum layer influences the heat transfer characteristics and thermal stability.

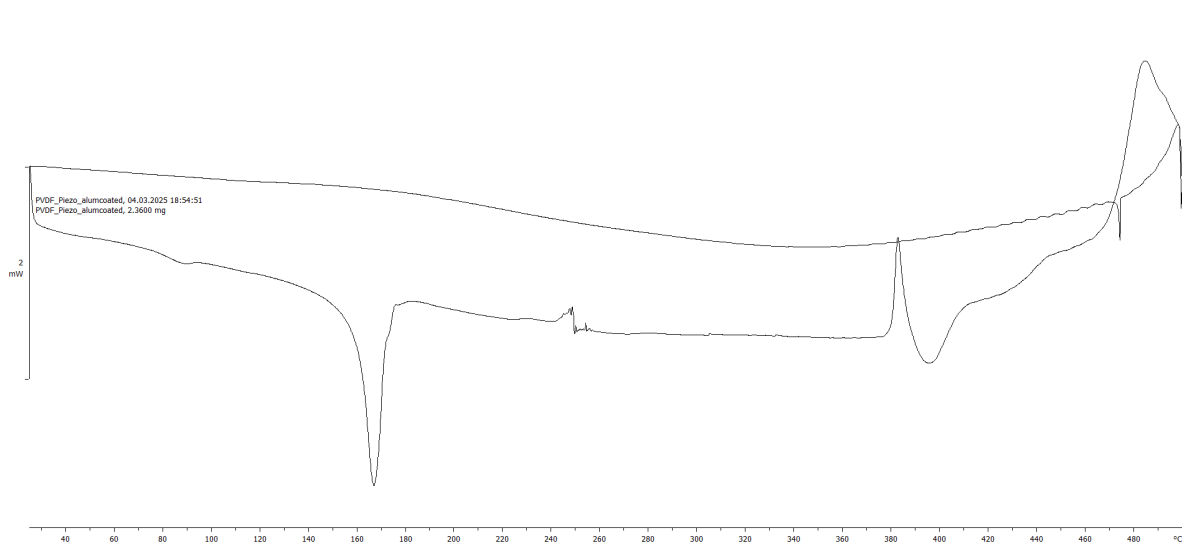


Figure 3.5: DSC thermograph of the aluminum-coated poled PVDF piezoelectric film.

Melting Behavior and Crystallinity: A prominent endothermic peak is observed at approximately 165–170°C, corresponding to the T_m of PVDF. This melting point is slightly lower than that of the non-coated PVDF film (175°C), which could indicate a minor reduction in temperature sensitivity due to the aluminum coating process. However, compared to the non-poled and non-coated samples, the area under the endothermic peak is significantly larger, indicating a higher enthalpy of fusion and suggesting an increased degree of crystallinity in the coated and poled PVDF. The presence of a well-defined, single-phase melting peak further supports the conclusion that aluminum-coating and poling promotes phase purity and enhances crystalline ordering in the material.

Additional Transitions and Aluminum Influence: In the DSC thermograph of the aluminum-coated, poled PVDF film, an unexpected exothermic peak is observed around 375°C, preceding the onset of the main thermal degradation event at approximately 420°C. This exothermic event is absent in both the non-coated poled and non-poled samples, suggesting that it may be attributed to the presence of the aluminum layer or to specific interactions between the coating and the polymer matrix. One possible explanation is that the aluminum coating facilitates a structural reorganization or oxidative cross-linking process within the PVDF prior to decomposition, releasing heat. Alternatively, the metal layer may act catalytically, altering the thermal decomposition pathway or stabilizing certain polymer segments until a delayed breakdown occurs. This unique thermal feature indicates that metal coating not only affects electrical and mechanical properties, but may also significantly influence the thermal behavior of the polymer.

Thermal Degradation: The degradation process is consistent with the other other samples, presenting a broad exothermic transition occurs between 420°C and 480°C.

3.2.4 DSC Conclusions

In the DSC analysis, the poled PVDF film exhibited a higher y-axis value—representing greater heat flow—during the melting transition compared to the non-poled sample. This observation aligns with the expected effect of electrical poling, which promotes the formation of the β -phase and enhances crystalline ordering within the polymer matrix. A higher heat flow peak indicates a greater enthalpy of fusion, which correlates with an increased degree of crystallinity. The sharper and more intense melting peak in the poled sample suggests that more energy is required to disrupt the crystalline regions, reflecting a more thermally stable and ordered structure. Therefore, the increased y-axis signal in the poled film is a meaningful indicator of improved crystallinity and phase uniformity induced by the poling process.

Overall, the analysis revealed distinct thermal behaviors among the three PVDF film samples—non-coated non-poled, non-coated poled, and aluminum-coated poled—highlighting the effects of poling and metal coating on crystallinity and thermal stability. The non-coated non-poled sample exhibited a

bimodal melting peak between 160°C and 180°C, indicative of a mixed crystalline phase composition, likely including both α - and β -phases. In contrast, the non-coated poled sample showed a single, sharp melting peak at 175°C, suggesting a more uniform β -phase composition and increased crystallinity induced by electrical poling. The aluminum-coated poled film displayed a similar sharp melting peak but exhibited the largest endothermic area among the three samples, indicating the highest degree of crystallinity. Additionally, it presented a unique exothermic peak around 375°C, absent in the other two samples. This feature suggests a potential interaction between the aluminum layer and the polymer matrix, possibly involving structural reorganization or catalytic effects prior to thermal degradation. Collectively, these results demonstrate that poling enhances phase purity and crystallinity, while metal coating can further increase crystalline content and significantly modify the high-temperature thermal response of PVDF.

3.3. Continuous and Cyclic Stretching to Mimic Heart Motion

The custom-designed mechanical stretching platform was developed to simulate the natural rhythmic contractions of cardiac tissue by applying cyclic uniaxial continuous deformation to cultured cardiomyocytes. This approach replicates the mechanical microenvironment of the native myocardium and study cell behavior under dynamic mechanical stimulation.

To achieve continuous and cyclic mechanical stimulation of the cell-laden PVDF membrane, we implemented an automated control protocol using an Arduino-based script [Figure 3.6]. The solenoid valve responsible for pneumatic actuation was connected to a digital output pin and controlled via a timing-based logic loop. The code alternates the solenoid state (open/close) at defined time intervals to produce rhythmic inflation and deflation of the stretch chamber, thereby mimicking cardiac-like cyclic strain.

```

1  // C++ code
2  //
3  int solenoidPin = 4;
4  unsigned long previousMillis = 0;
5  unsigned long interval = 10000;
6  bool state = false;
7  unsigned long startTime;
8
9
10 void setup() {
11     pinMode(solenoidPin, OUTPUT);
12     startTime = millis();
13 }
14
15
16 void loop() {
17     unsigned long currentMillis = millis();
18     unsigned long elapsed = currentMillis - startTime;
19
20
21     // Ramp protocol based on elapsed time
22     if (elapsed < 2 * 3600000) {
23         interval = 5000; // 0-2 hrs → 0.2 Hz
24     } else if (elapsed < 4 * 3600000) {
25         interval = 2000; // 2-4 hrs → 0.5 Hz
26     } else {interval = 1000; // 4-24 hrs → 1 Hz}
27
28
29
30     if (currentMillis - previousMillis >= interval / 2) {
31         previousMillis = currentMillis;
32         state = !state;
33         digitalWrite(solenoidPin, state ? HIGH : LOW);
34     }
35 }
36

```

Figure 3.6: Arduino control code for cyclic actuation of the solenoid valve. The script generates continuous and automated stretching by toggling the solenoid state at defined intervals.

The implemented control algorithm initiates a ramped stimulation protocol that gradually increases the frequency of mechanical deformation over time. Specifically, the code sets an initial actuation interval of 5 seconds (0.2 Hz) for the first 2 hours, followed by 2-second intervals (0.5 Hz) from 2 to 4 hours, and

finally stabilizes at 1-second intervals (1 Hz) for the remainder of the 24-hour period. This progressive ramping strategy was intentionally designed to prevent sudden mechanical stress that could lead to detachment of adherent cells from the PVDF film. By gradually increasing the cyclic frequency, the system allows cells to acclimate to mechanical stimulation in a controlled manner, thereby enhancing cell retention and viability on the membrane surface.

This programmable, frequency-ramped control logic ensures that the device delivers physiologically relevant cyclic stretching continuously and without interruption. Moreover, the automated timing loop supports experimental reproducibility and eliminates the need for manual intervention, fulfilling the design goals of standardization and operational reliability essential for long-term *in vitro* studies.

3.4. Compact Design Compatible with Standard Incubators for Parallel Experimental Throughput

The entire mechanical stimulation platform was designed to be compact and modular, enabling its integration into a standard cell culture incubator without modification of environmental controls. This allows experiments to be conducted under stable temperature, CO_2 , and humidity conditions, which are crucial for maintaining cardiomyocyte viability.

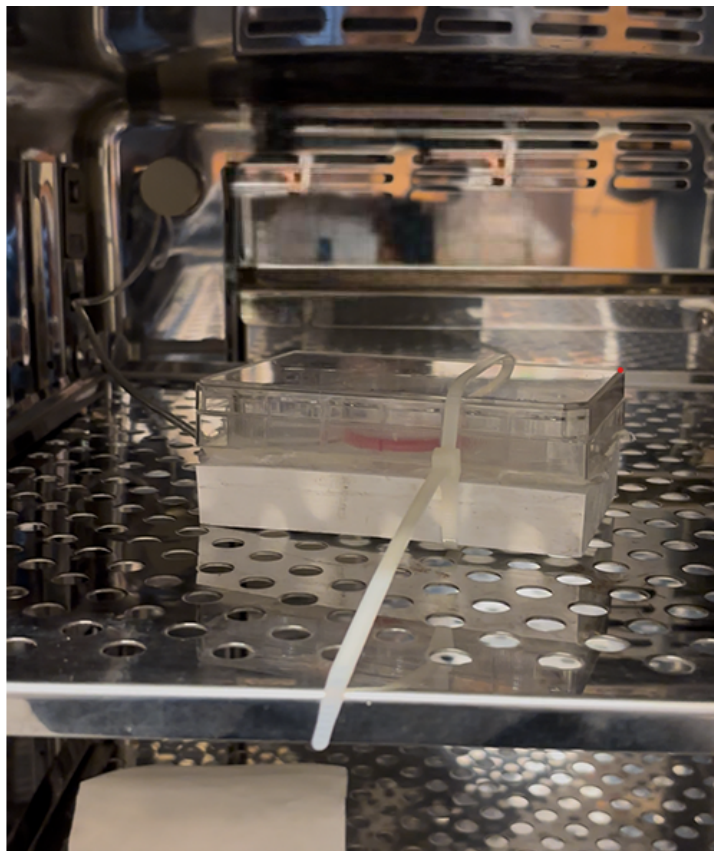


Figure 3.7: Custom stretching device placed inside a standard cell culture incubator. This setup demonstrates compatibility for long-term operation under physiological conditions, enabling continuous mechanical stimulation of cultured cells.

The device can support parallel testing by adapting multiple modified wells within a standard 6-well plate format. While only one well was used in this study due to cell availability constraints, the system is scalable and allows simultaneous stimulation of multiple wells under identical or variable conditions. This feature provides the flexibility to compare different material substrates, strain regimens, or pharmacological treatments within a single experimental run, improving reproducibility and throughput.

3.5. Generate Physiologically Relevant Stress and Strain Levels

Initial testing demonstrated that the PDMS membrane began to inflate at a minimum flow rate of 20 mL/s. Using the flow parameters and Equation 2.3, the air velocity through the tube was estimated at approximately 25.5 m/s, corresponding to a dynamic pressure of 397 Pa [Equation 2.1]. At this pressure, the membrane deformed into a semi-spherical shape, indicating successful mechanical actuation. To estimate the displacement volume produced during PDMS membrane inflation, the deformation was modeled as a spherical cap. The vertical amplitude (h) of the inflated membrane and the base radius (r) of the membrane were measured to be 1 mm and 5 mm, respectively. The displaced volume (V) of the dome-shaped inflation was calculated using the spherical cap formula:

$$V = \frac{1}{6}\pi h (3r^2 + h^2) \quad (3.1)$$

Substituting the values:

$$V = \frac{1}{6}\pi(1) [3(5)^2 + (1)^2] = \frac{1}{6}\pi(75 + 1) = \frac{1}{6}\pi(76) \quad (3.2)$$

$$V \approx \frac{76\pi}{6} \approx 39.8 \text{ mm}^3 \quad (3.3)$$

This displacement volume indicates the amount of air required to generate a 1 mm vertical deformation of the PDMS membrane, which directly applies mechanical strain to the PVDF film and indirectly to the cultured cells.

3.5.1 Measured Strain Output of the PVDF Stretch System

The strain applied to the PVDF film during cyclic actuation was measured using the method described in Section 2.2.5.2. The initial marker distance was 0.11 cm in the relaxed state, and 0.12 cm in the fully stretched state. This corresponds to:

$$\text{Strain} = \frac{0.12 - 0.11}{0.11} = \frac{0.01}{0.11} \approx 0.0909 \text{ or } 9.1\% \quad (3.4)$$

This value falls within the physiological strain range of 5–15% experienced by cardiomyocytes during contraction, as supported by prior biomechanical studies. Achieving this target range indicates that our stretch system successfully replicates the magnitude of mechanical strain found in native cardiac tissue, thereby validating the mechanical functionality of our device for generating piezoelectric voltage stimulation in *in vitro*.

These results confirm that the stretch system is responsive to low pneumatic pressures and capable of producing the desired membrane deformation. The observed behavior validates the system's functionality and its potential to simulate cardiac mechanical strain for further electromechanical investigations.

3.6. Device Assembly

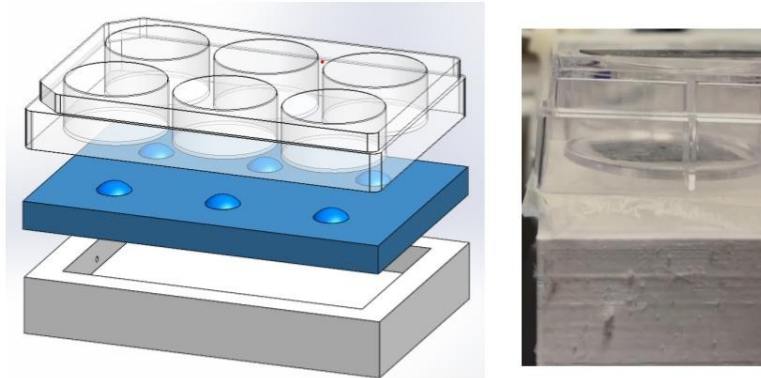


Figure 3.8: Assembly of the custom cyclic stretch system designed to interface with a standard 6-well plate.

The final assembly of the stretch system integrated all mechanical, electronic, and biological components into a compact, incubator-compatible platform capable of sustaining cyclic mechanical stimulation. Central to the setup was a custom-fabricated PDMS membrane, which served as the flexible base for a multi-well plate allowing direct cell culture. This membrane was bonded to a 3D-printed chamber designed to withstand repeated pressurization cycles while maintaining sterility and ease of access. Above the PDMS layer, the cell culture plate was securely affixed to ensure uniform strain transmission across all wells [Figure 3.8].

The actuation system consisted of a solenoid valve controlled via an Arduino microcontroller, which was programmed with a ramped frequency protocol to gradually increase the actuation rate from 0.2 Hz to 1 Hz over 24 hours. The solenoid modulated airflow from a pressurized source into the chamber beneath the PDMS, generating consistent, cyclic upward deformation of the membrane—mimicking cardiac contractility. All electronic components, including the power supply, control circuitry, and valve interface, were housed externally to preserve sterility and maintain incubator compatibility [Figure 3.9]. This fully integrated platform demonstrated reliable, automated cyclic stretching of the PVDF films, establishing a physiologically relevant testbed for future biological evaluation of cell responses to piezoelectric stimulation.



Figure 3.9: Arduino control circuit and solenoid valve were situated outside the incubator to maintain sterile culture conditions and prevent thermal or humidity-related damage to electronics.

Conclusion and Discussion

This project set out to address a pressing unmet need in cardiac regenerative medicine: the lack of minimally invasive, effective technologies that can stimulate cardiac tissue repair following myocardial infarction. Our goal was to successfully improve experimental reproducibility through automation and standardization by designing, constructing, and validating a stretch system that enables *in vitro* characterization of piezoelectric PVDF films under physiological-like conditions.

One of the major challenges in evaluating the regenerative potential of motion-induced electrical stimulation lies in replicating the dynamic, cyclic nature of cardiac contraction in a controlled lab environment. We addressed this by engineering a programmable stretch system capable of mimicking physiological heartbeats through solenoid-driven inflation and deflation. A custom Arduino-based code was implemented to drive the solenoid in a cyclic manner, with ramped frequency adjustments over time. This ramping protocol—starting at 0.2 Hz, increasing to 0.5 Hz after 2 hours, and reaching 1 Hz after 4 hours—was intentionally designed to mimic the progressive adaptation of cardiomyocytes to mechanical stimulation. By gradually increasing the frequency, we aimed to reduce the risk of cellular detachment and enhance cell tolerance to stretch-induced strain. This approach ensured a continuous and cyclic motion profile that not only simulated *in vivo* cardiac conditions but also supported cellular adherence and survival over extended periods of mechanical stimulation.

The stretch system is essential for assessing the electromechanical behavior of the PVDF film and will serve as the basis for future biological assays. Thermal and mechanical characterization confirmed the biocompatibility and strain-responsiveness of the material, while inflation-deflation testing validated the consistency and repeatability of the cyclic strain profile. These achievements directly align with our initial goals: (1) to demonstrate voltage generation from cyclic strain, (2) to develop a reliable platform for *in vitro* cardiomyocyte stimulation, and (3) to lay the foundation for future bioassays assessing tissue regeneration.

Due to limited timing and restricted access to hiPSC-derived cardiomyocytes during the early phases of the project, we initially conducted biological testing using primary mouse myoblasts. This enabled early-stage evaluation of biocompatibility with PVDF films and adhesives while waiting for cardiomyocyte availability. Once we received the hiPSC-cardiomyocytes, we proceeded with static and stretch experiments, but the limited cell quantity significantly impacted experimental design. In our initial cardiomyocyte trial, we divided 1.5 million cells across three wells, leading to poor confluency and insufficient intercellular connections for spontaneous beating. In a subsequent attempt, we seeded all 1.5 million cells into a single well with polarized PVDF to improve local density. While attachment was improved, the large surface area of the 6-well plate still prevented the formation of a fully connected syncytium. For future experiments, scaling down to smaller well formats such as 24-well plates would increase local cell density, facilitate replicate testing, and improve the likelihood of observing contractile behavior and cardiomyocyte maturation.

Despite the success of the system’s mechanical function and early biocompatibility testing, the project was constrained by time and material availability. While the mechanical stretch system achieved reliable deformation of the PVDF film, additional refinements in automation, strain quantification, and long-term culture compatibility will be necessary for future experiments. These developments will support more rigorous *in vitro* validation and eventual *in vivo* testing.

Beyond the success of this particular device, our work contributes meaningfully to the broader field of cardiac repair technologies. The development of a self-powered, bioactive implant holds significant promise for the next generation of minimally invasive therapeutic interventions. Unlike external stimulation systems, our piezoelectric device eliminates the need for batteries or transcutaneous wires, reducing infection risks and improving long-term implantability. Moreover, the validated stretch system provides a critical *in vitro* testing platform that can be used for future studies on electrical stimulation thresholds, piezoelectric material optimization, and cardiomyocyte behavior under different mechanical conditions.

In conclusion, this project successfully met its core objectives and established a scalable, modular system for evaluating piezoelectric cardiac implants. Our findings support the feasibility of a motion-driven stimulation platform and open the door to continued innovation in cardiac regenerative therapies. Future directions include integrating real-time voltage monitoring, expanding to co-culture systems, and ultimately moving toward *in vivo* validation of the implant. By bridging materials science, bioengineering, and regenerative medicine, our work represents a promising step forward in addressing the burden of cardiovascular disease through novel, energy-autonomous biomedical devices.

Future Prospects

Future work should address current limitations and build upon the foundational work established here to enable more rigorous biological validation. A focus on completing biological validation of the stretch system using hiPSC-derived cardiomyocytes under dynamic strain is necessary. One major improvement would be to transition from a 6-well plate to a smaller format, such as a 24-well. This change would allow for a higher seeding density, which is critical for cardiomyocytes to establish strong connections with each other and begin beating in a coordinated manner. A smaller well format would also support multiple experimental replicates within the same run, improving the reliability and reproducibility of results.

Another key recommendation is to delay the onset of mechanical stretching until after the cardiomyocytes begin to beat spontaneously. Starting stretch once contractions are visible would make it possible to assess how cyclic mechanical strain and piezoelectric stimulation influence beating frequency and overall functional maturation. Additionally, integrating a real-time monitoring system—such as video tracking to detect beating or sensors to confirm strain, is essential to ensure the device is functioning properly throughout the culture period.

Beyond *in vitro* biological studies, a critical next step is the direct measurement and characterization of the voltage output generated by the PVDF film during mechanical deformation. While the current system was validated for its ability to maintain cyclic strain in a biologically compatible environment, understanding the electromechanical behavior of the material is essential for quantifying the electrical stimuli experienced by the cells. Integrating electrodes and a signal amplification circuit would allow real-time monitoring of voltage generation across strain cycles, enabling more precise tuning of stimulation parameters and providing quantitative insight into the relationship between strain amplitude, frequency, and piezoelectric output. Such data are necessary for optimizing the platform as an electromechanical stimulation device and for tailoring it to specific regenerative or electrophysiological applications.

Furthermore, it would be beneficial to repeat the DSC experiments using a broader temperature range, ideally from -100°C to 500°C . This extended range would allow for the accurate identification of the glass transition temperature (T_g) of PVDF, which typically occurs around -35°C but may shift depending on processing conditions, crystallinity, and phase composition. Capturing the T_g is essential for fully characterizing the thermal behavior of the material, particularly for applications where low-temperature mechanical flexibility or dielectric performance is relevant. Additionally, a wider range would ensure that all relevant thermal events—such as relaxation processes, sub- T_g transitions, and the full degradation profile—are thoroughly observed and analyzed.

In parallel, the stretch platform could be expanded to test other piezoelectric materials, optimize stimulation parameters, or support different tissue models beyond the heart. Applications in skeletal muscle regeneration, tendon mechanobiology, or neuromuscular engineering may benefit from this adaptable and programmable stimulation platform.

Ultimately, the stretch system developed in this project serves as a foundation for future *in vitro* studies that can inform the development of implantable, self-powered therapeutic devices. Collectively, these improvements would transform the stretch device into a powerful tool for studying and developing self-powered biomedical therapies.

6

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A

Appendix



Figure A.1: View from the top of the final PDMS layer.



Figure A.2: 3D printed PLA bottom chamber for the stretch system.



Figure A.3: 6-well plate with 15 mm and 10 mm holes at the bottom.

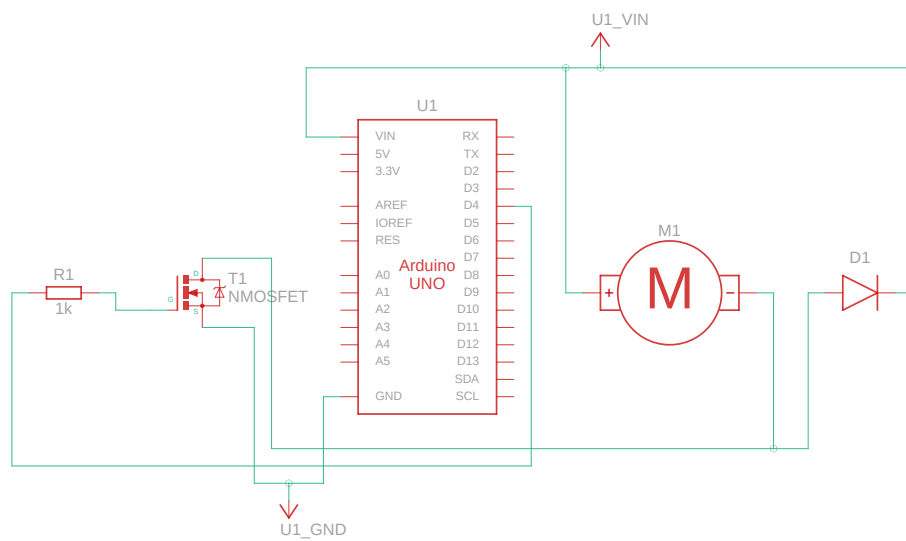


Figure A.4: Circuit diagram for automated solenoid valve control using an Arduino Uno R4 and MOSFET. This hardware configuration enables digital control of pneumatic actuation at 1 Hz, facilitating cyclic mechanical stimulation in the experimental setup. (Chris @ BCR, 2015)